

# DAY 01 — EXCEL + DATA FOUNDATION

## Technical Analyst Learning Notes

### 1. Learning Objective

Build the foundation required before serious data analysis: understand the dataset, identify its grain, check data quality and data types, distinguish dimensions and measures, structure data correctly in Excel, and calculate basic sales KPIs correctly.

### 2. Core Analyst Workflow

- Understand the business objective
- Identify the dataset grain
- Understand columns and data types
- Check data quality
- Identify dimensions and measures
- Define the KPI
- Calculate
- Validate the calculation
- Interpret the result
- Identify what needs further investigation

**Golden principle: Do not start calculating before understanding the data.**

### 3. Dataset Fundamentals

| Concept        | Meaning                                                         |
|----------------|-----------------------------------------------------------------|
| Dataset        | Collection of related records and attributes used for analysis. |
| Record / Row   | One individual observation or transaction.                      |
| Field / Column | An attribute describing each record.                            |
| Data Grain     | Exact level of detail represented by one row.                   |

**Our sales dataset grain: ONE BOOKING TRANSACTION / BOOKING RECORD.**

Why grain matters: it tells us what each row represents, prevents incorrect aggregation, clarifies what analysis is possible, and prevents mixing levels of detail.

### 4. Data Types

- Text — country, branch, salesperson, product, customer, booking ID.
- Date — booking date.
- Number — pax count, sales amount, target amount.
- Boolean / Yes-No — TRUE/FALSE where applicable.

Data type and formatting are different. A value can look like a number but be stored as text.

## 5. Data Quality

- Missing values / blanks
- Duplicate records
- Invalid values
- Numbers stored as text
- Incorrect data types
- Negative or unusual values
- Inconsistent text values
- Unexpected values / potential outliers

**Investigation process: IDENTIFY → INVESTIGATE → VALIDATE → FIX**

Never blindly delete or change questionable data; first establish whether it is genuinely wrong, missing, or simply unusual.

## 6. Dimensions vs Measures

| Type      | Examples                                        | Purpose                     |
|-----------|-------------------------------------------------|-----------------------------|
| Dimension | Country, Branch, Salesperson, Product, Customer | Slice / categorize analysis |
| Measure   | Pax, Sales Amount, Target Amount                | Calculate / aggregate       |

## 7. Excel Foundation

- Workbook — Excel file.
- Worksheet — individual sheet.
- Cell — intersection of row and column.
- Range — collection of cells.
- Row — horizontal set of cells; in a sales table normally one record.
- Column — vertical set of cells; normally one attribute.

Good raw-data structure:

- One row = one record
- One column = one attribute
- One clear header row
- No unnecessary merged cells
- Consistent data types within a column
- Do not mix subtotals/presentation formatting into raw data

## 8. Excel Table

Sales dataset table name: tbl\_sales

- Built-in filtering
- Automatic expansion when rows are added
- Structured references
- Easier formulas
- More reliable analysis than an unmanaged range

Example structured reference:  $=[@\text{Sales\_Amount}]/[@\text{Target\_Amount}]$

## 9. KPI Fundamentals

| Term     | Meaning                                                              |
|----------|----------------------------------------------------------------------|
| Metric   | A measurable business value.                                         |
| Measure  | A value calculated or aggregated for analysis.                       |
| KPI      | A key performance indicator tied to an important business objective. |
| Target   | Expected or desired level of performance.                            |
| Variance | Difference between actual and target.                                |

Not every metric is a KPI. A KPI is important for evaluating a defined business objective.

## 10. Actual, Target, Achievement & Variance

**Achievement % = Actual Sales ÷ Target Sales × 100**

**Variance = Actual Sales – Target Sales**

- Positive variance = actual above target.
- Negative variance = actual below target.
- Zero variance = actual equals target.

**Overall Achievement % = Total Actual Sales ÷ Total Target Sales × 100**

Do not automatically average row-level achievement percentages. A simple average gives every row equal weight even when target values differ.

## 11. Excel Formula Fundamentals

- Formulas begin with =
- $=A2+B2$
- $=B2-C2$
- $=B2/C2$

**Achievement %:**  $=[@\text{Sales\_Amount}]/[@\text{Target\_Amount}]$

**Variance:**  $=[@\text{Sales\_Amount}]-[@\text{Target\_Amount}]$

**Total Sales:**  $=\text{SUM}(\text{tbl\_sales}[\text{Sales\_Amount}])$

**Total Target:**  $=\text{SUM}(\text{tbl\_sales}[\text{Target\_Amount}])$

**Overall Achievement:**  $=\text{SUM}(\text{tbl\_sales}[\text{Sales\_Amount}])/\text{SUM}(\text{tbl\_sales}[\text{Target\_Amount}])$

## 12. Relative vs Absolute References

- Relative reference: changes when copied. Example:  $=A2*B2$
- Absolute reference: remains fixed when copied. Example:  $=A2*\$B\$1$

**Memory rule: Relative = moves. Absolute = stays.**

### 13. Basic Excel Functions Introduced

- SUM — adds values.
- AVERAGE — arithmetic average.
- MIN — smallest value.
- MAX — largest value.
- COUNT — counts numeric cells.
- COUNTA — counts non-empty cells.

### 14. Fact → Finding → Hypothesis → Investigation → Action

| Term          | Meaning                                               |
|---------------|-------------------------------------------------------|
| Fact          | What the data directly establishes.                   |
| Finding       | A meaningful observation revealed by analysis.        |
| Hypothesis    | A possible explanation that still needs validation.   |
| Investigation | Additional analysis/data used to test the hypothesis. |
| Action        | Next business step based on validated information.    |

- Fact: Overall achievement is 95%.
- Finding: Six bookings are below target.
- Hypothesis: A country/branch/product/salesperson may contribute to the shortfall.
- Investigation: Break performance down by relevant dimensions and time.

### 15. Analytical Question Thinking

A vague statement such as “Nigeria sales are low” should be converted into specific questions:

- Low compared with what — target, previous period, or another benchmark?
- What time period — month, quarter, year, or YTD?
- When did the decline begin?
- Is it a one-time fluctuation or a sustained trend?
- Is it specific to Nigeria or visible elsewhere?
- Is the issue revenue, bookings, achievement, or another measure?
- Is the issue at country, branch, product, or salesperson level?

### 16. Day 01 Practical Results

| Analysis              | Result                 |
|-----------------------|------------------------|
| Total Sales           | 53,200                 |
| Total Target          | 56,000                 |
| Overall Achievement   | 95.00%                 |
| Highest Achievement % | B010                   |
| Lowest Achievement %  | B006                   |
| Bookings above 100%   | B005, B007, B009, B010 |

## 17. Day 01 Final Assessment — Key Answers

| Question         | Key Answer                                                                                    |
|------------------|-----------------------------------------------------------------------------------------------|
| Q1 — Grain       | One booking record / booking transaction.                                                     |
| Q2 — Row         | One booking transaction.                                                                      |
| Q3 — Dimensions  | Country, Branch, Salesperson, Product, Customer.                                              |
| Q4 — Measures    | Pax, Sales Amount, Target Amount.                                                             |
| Q5 — Achievement | $\text{Sales Amount} \div \text{Target Amount} \times 100$ .                                  |
| Q6 — Variance    | $\text{Actual Sales} - \text{Target Sales}$ .                                                 |
| Q7 — Overall KPI | $\text{Total Actual} \div \text{Total Target}$ ; do not simply average row-level percentages. |
| Q8 — References  | Relative changes when copied; absolute remains fixed.                                         |
| Q9 — Example     | Actual 8,000; Target 10,000 → Variance -2,000; Achievement 80%.                               |

## 19. Golden Rules

1. Understand the grain before analyzing.
2. Check data quality before trusting results.
3. Know what each column represents.
4. Separate dimensions from measures.
5. Define a KPI before calculating it.
6.  $\text{Variance} = \text{Actual} - \text{Target}$ .
7.  $\text{Overall Achievement} = \text{Total Actual} \div \text{Total Target}$ .
8. Do not confuse a hypothesis with a proven cause.
9. Validate calculations, not just formulas.
10. Do not jump into advanced analysis before understanding the data.